# Slick Trace investigation report — case_0005

Generated 2026-09-26T12:21:41.115679+00:00 · reproducible investigation report

**What this document is not.** This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

## Finding

**Leading candidate: 419420729**. This is a ranking by physical consistency under tested assumptions — an investigative lead for a human analyst, not proof of responsibility.

## Scene

| Field | Value |
|---|---|
| Incident | case_0005 |
| Acquired (UTC) | 2026-09-10T04:30:00+00:00 |
| Sensor | synthetic SAR-like (VV proxy, single channel) |
| Data status | synthetic |
| Pixel spacing | 433.59 m |
| Label source | generator ground truth (synthetic) — not a human or independently confirmed label |

## Evidence layers

| Layer | Provenance |
|---|---|
| AIS | synthetic (fictional MMSIs; no real broadcast data) |
| Forcing | slicktrace_core.synth_generator constant+spatial field v0.2 |
| Licence | Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply. |

## Traffic filtering

| State | Vessels |
|---|---|
| candidate | 4 |
| unassessed | 1 |
| excluded | 1 |

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

## Rank fragility

Leader retained in 0.825 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

## File integrity

| File | SHA-256 (first 16) |
|---|---|
| attribution_result.json | 1ce44b82fc967942 |
| coverage_result.json | e16cb40e5a418dee |
| forcing.npz | 34b71c9eccb0e5c6 |
| forecast_result.json | f9decb564b65b8d2 |
| fragility_result.json | 12669ef777c21611 |
| historical_mapping.json | 927931707c12bdd4 |
| manifest.json | bca1d457d1e1bad0 |
| mask_groundtruth.png | 0ea1184f6549eef2 |
| naive_baseline_result.json | 470ff4ec0ba05f46 |
| next_observation_result.json | 6f6f1e25763cbd90 |
| scene.png | 39be2f9e4d7a757f |

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

## Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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